Discover Public Health

RESEARCH Open Access

Knowledge, attitude and practice toward blood donation and associated factors among college students in Jumla, Nepal

Deependra Muraw[1†], Nisha Adhikari[1†], Pradeep Narayan Joshi[1] and Dirgha Raj Joshi[1*]

[†]Deependra Muraw and Nisha Adhikari have contributed equally to this work.

*Correspondence:
Dirgha Raj Joshi
djmeropaila121@gmail.com
[1]School of Pharmacy, Karnali Academy of Health Sciences, Jumla, Nepal

Abstract

Background Blood donation is a vital component of modern healthcare systems, as blood cannot yet be fully synthesized for routine clinical use and depends entirely on voluntary human donors. College students represent a large and healthy pool of potential donors; however, evidence regarding their knowledge, attitudes, and donation practices in geographically remote regions such as Jumla, Nepal, remains limited.

Objective This study aimed to assess the knowledge, attitude, and practice (KAP) of blood donation and identify associated factors among college students in Jumla, Nepal.

Methods A cross-sectional analytical study was conducted among 647 students from three educational institutions in Jumla: Karnali Academy of Health Sciences (KAHS), Karnali Technical School (KTS), and Jumla Multiple Campus (JMC). Participants were selected using an institution-based stratified convenience sampling approach and completed a structured self-administered questionnaire. Data were analyzed using descriptive statistics and Chi-square tests with SPSS version 23.

Results Among the 647 participants, 73% demonstrated a positive attitude toward blood donation and 82.7% expressed willingness to donate voluntarily. However, only 19.5% had ever donated blood. Overall knowledge levels were generally low, with only 19.3% achieving a good knowledge score. Blood donation practice was significantly associated with age, gender, academic stream, and year of study ($p < 0.05$). The most commonly reported barrier to donation was lack of opportunity (44.7%), followed by medical reasons and fear of needles.

Conclusion Despite favorable attitudes and strong willingness toward voluntary blood donation, actual donation practice among college students in Jumla remained limited. The findings suggest that limited donation opportunities and inadequate awareness may contribute to the gap between intention and practice. Regular campus-based blood donation campaigns, targeted educational interventions, and strategies addressing fear-related concerns may help improve voluntary blood donation participation among students.

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

Highlights

- Out of 647 valid responses, most college students in Jumla showed positive attitudes (73%) and strong willingness to donate blood (82.7%).
- Despite favorable perceptions, actual donation practice was low (19.5%), indicating a clear knowledge–practice gap.
- Lack of donation opportunities and limited local infrastructure were the major barriers to blood donation among students

Keywords Blood donation, Knowledge, attitude, and practice, College students, Blood donation awareness, Voluntary non-remunerated blood donation

1 Introduction

Blood donation (BD) is a process where a healthy person voluntarily gives blood that is collected, screened, processed, and stored for various life-saving components, such as red cells, platelets, and plasma [1, 2]. Despite significant technological advances in medicine, blood cannot yet be manufactured for routine clinical use, and healthcare systems therefore continue to rely entirely on voluntary human donors for transfusion support [3, 4]. Its clinical necessity plays a key role in today's medical care, essential for life-saving conditions like traumatic injuries, surgery, long-term illnesses, or heavy bleeding during childbirth [1, 5]. Globally, the World Health Organization (WHO) reports approximately 118.5 million annual blood donations. However, a significant gap exists between high-income and low-income countries. In 79 countries, over 90% of the blood supply comes from voluntary, unpaid donors, but 54 countries still depend on family donors, replacement donors, or paid donors [6].

The global perspective on donor motivation shows a complex landscape of altruism, rewards, and systemic challenges [7]. Although the WHO advocates for 100% voluntary non-remunerated blood donation (VNRBD), many regions still struggle with replacement donation, where family members must provide blood for a specific patient [8, 9]. In some scenarios, appreciation through certificates or small tokens is used to encourage donors, but the persistence of paid donation in some parts of the world poses significant risks to blood safety and ethics [6, 10]. Challenges in maintaining an adequate blood supply include entrenched misconceptions, fear of needles (trypanophobia), and limited infrastructure, particularly in "blood deserts"- geographical areas where local demand for blood cannot be met in at least 75% of emergency cases [7, 11].

Youth and college students represent an important donor pool globally, as they represent a large, healthy, and accessible pool of potential long-term donors [12, 13]. Studies indicate that while students often possess positive attitudes toward donation, their actual practice remains suboptimal due to misconceptions, needle phobia, limited awareness, or the simple fact of never being asked [9, 14, 15]. In Nepal, blood transfusion services are primarily coordinated through a national network managed by the Nepal Red Cross Society in collaboration with hospital-based laboratories, operating 121 blood centers across 76 districts. Despite this broad coverage, a significant gap still persists between blood supply and demand [1, 7, 13, 16]. Approximately 90% of blood donations in Nepal are voluntary non-remunerated, while replacement donation still contributes to the remaining supply in certain settings [8]. Despite this extensive system, services remain largely concentrated in urban areas, where infrastructure and transfusion capacity are more developed, whereas geographically remote and mountainous districts often

experience uneven distribution of services and limited institutional capacity [16]. In remote districts such as Jumla, limited blood transfusion infrastructure and absence of fully established or permanently accessible blood bank services may restrict timely availability of blood during emergencies, particularly in geographically constrained health service settings [17]. While knowledge, attitude, and practice (KAP) studies are available for urban areas such as Kathmandu and Chitwan, evidence from remote high-mountain regions like Jumla remains limited [3, 4]. Identifying factors associated with blood donation behavior among students is essential for designing targeted interventions to strengthen voluntary blood donation systems. Therefore, this study was conducted to assess and compare the KAP of blood donation among college students in Jumla, Nepal, providing the first baseline data for this remote region and exploring the potential of youth and college students as a healthy donor pool to strengthen the local blood supply.

2 Materials and methods

2.1 Study design and setting

A cross-sectional study was conducted among college students in Jumla, Nepal, at three major institutions representing diverse academic streams: Karnali Academy of Health Sciences (KAHS) for health sciences, Karnali Technical School (KTS) for technical and vocational training, and Jumla Multiple Campus (JMC) for education and management programs. These institutions encompass the main categories of higher education in the district.

2.2 Study population and eligibility

The study population included college students enrolled in the selected institutions, representing diverse academic programs classified into two streams: Health Sciences (Bachelor of Pharmacy [B.Pharm], Bachelor of Public Health [BPH], Master of Public Health [MPH], Bachelor of Science in Nursing [B.Sc. Nursing], Bachelor of Medicine and Bachelor of Surgery [MBBS], Bachelor of Midwifery Science [BMS], Bachelor of Nursing Science [BNS], Health Assistant [HA], Diploma in Pharmacy [D.Pharm]) and Non-Health Sciences (Diploma in Civil Engineering [DCE], Diploma in Agriculture [D.Ag], Bachelor of Education [B.Ed.], Bachelor of Business Studies [BBS]). Students aged 18 years and above who were present and willing to participate were included.

2.3 Sampling technique and participant recruitment

Participants were recruited using an institution-based stratified convenience sampling approach. The three selected institutions - KAHS, KTS, and JMC- served as sampling strata. Eligible students were approached during scheduled classes and breaks. A total of 720 questionnaires were distributed, of which 704 were returned, yielding a response rate of 97.8%. Among the returned questionnaires, 57 were excluded due to incomplete or missing responses. Finally, 647 valid questionnaires were included in the final analysis.

2.4 Study instrument

A structured, self-administered questionnaire was adapted from validated KAP tools and modified for the local Nepalese context (Neupane et al. [3]; Kittu et al. [2]; Salem et al. [18]). It was pre-tested among 20 students to ensure clarity and cultural appropriateness. The questionnaire included four sections: socio-demographics (age, gender, institution,

academic stream, year of study); knowledge (15 questions on eligibility, requirements, and benefits); attitude (9 Likert-scale items on safety, altruism, and fear); and practice (8 items; previous donation history, frequency, and reasons for or against donation).

2.5 Data collection procedure

Data were collected with the assistance of trained research assistants. Participants were briefed on the study's objectives and the voluntary nature of their participation. Before data collection, written informed consent was obtained from all participants. To ensure privacy and encourage honest responses, the questionnaires were completed anonymously.

2.6 Variable scoring

Participant responses were scored using Modified Bloom's cut-off points. In the Knowledge section, correct answers received 1 point and incorrect or "unsure" responses 0, with total scores classified as Good (80–100%), Moderate (60–79%), or Poor (< 60%). Attitude was assessed on a 3-point scale (Agree, Neutral, Disagree), with higher scores indicating a more positive outlook, using the same percentage-based thresholds. Modified Bloom's cut-off points are commonly applied in KAP studies for meaningful categorization of knowledge and attitude levels (Kaliyaperumal, 2004) [19].

2.7 Data analysis

Data were cleaned, coded, and entered into Microsoft Excel 2019, then analyzed using SPSS version 23. Descriptive statistics (frequencies, percentages) summarized demographic and KAP data, while Chi-square tests were used to assess associations, with $p < 0.05$ considered statistically significant.

3 Results

3.1 Demographic profile

Most participants were young adults, female, and enrolled in Health Sciences programs. Second-year students constituted the largest proportion of respondents. Detailed demographic characteristics are presented in Table 1.

Table 1 Demographic characteristics of the participants ($n = 647$)

Demographic characteristics	Category	%
Age	18–22	71.6
	23–26	21.2
	27–30	4.9
	> 30	2.3
Gender	Male	43.9
	Female	56.1
Institute	KAHS	43.1
	KTS	34.0
	JMC	22.9
Study year	1st Year	29.7
	2nd Year	50.5
	3rd Year	11.3
	4th Year	8.5
Academic stream	Health Sciences	63.1
	Non-Health Sciences	36.9

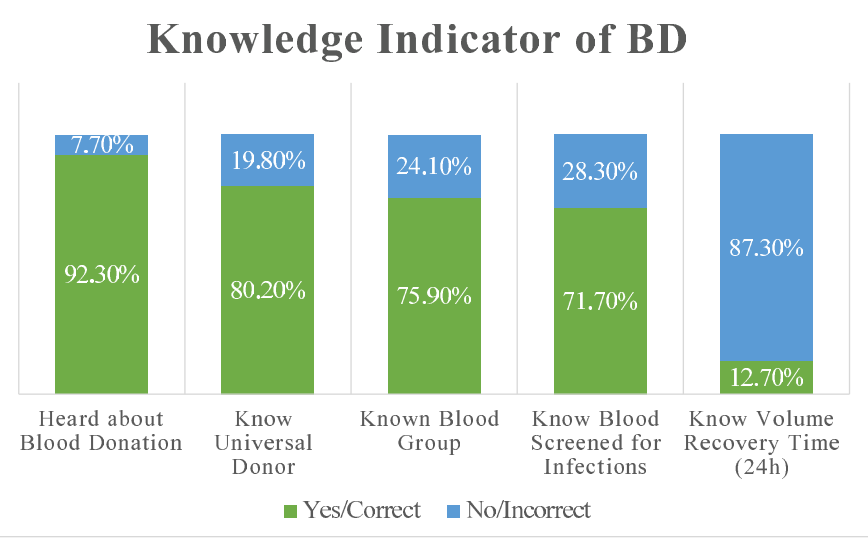


Fig. 1 Knowledge indicator of blood donation ($n = 647$)

Table 2 Association between demographic characteristics and knowledge level of blood donation ($n = 647$)

Variables	Knowledge level category, %			*p*-value
	Poor	Moderate	Good	
Age group (years)				
18–22	51.8	35.9	12.3	< 0.001*
23–26	25.5	36.5	38.0	
27–30	21.9	56.3	21.9	
> 30	6.7	33.3	60.0	
Gender				
Male	43.0	37.7	19.4	0.93
Female	44.4	36.4	19.3	
Academic stream				
Health sciences	30.1	43.4	26.5	**< 0.001***
Non-health sciences	67.0	25.9	7.1	
Year of study				
1st year	47.9	35.9	16.1	**< 0.001***
2nd year	47.7	37.6	14.7	
3rd year	35.6	32.9	31.5	
4th year	16.4	41.8	41.8	
Overall	**43.7**	**36.9**	**19.3**	

*Statistically significant at $p < 0.05$

3.2 Knowledge level regarding blood donation

Most participants had heard about blood donation and demonstrated basic awareness regarding blood groups and blood screening procedures. However, overall knowledge remained suboptimal, particularly regarding technical aspects such as post-donation blood volume recovery. Good knowledge levels were significantly associated with age, academic stream, and year of study, with higher scores observed among older students, senior-year students, and Health Sciences students ($p < 0.001$) (Fig. 1; Table 2).

Table 3 Attitudes and perceptions toward blood donation ($n = 647$)

Variable	Agree %	Neutral %	Disagree %
BD is a noble act and social duty	89.0	7.9	3.1
Willing to donate voluntarily	82.7	11.4	5.9
Encourage others to donate	89.3	7.6	3.1
Overcoming fear of needles matters	34.6	29.8	35.6
Organizing regular BD camps at the college	77.7	17	5.3
Feels proud when donating blood	82.2	14.4	3.4
Women can donate blood as safely as men	71.1	20.1	8.8
Promote BD in rural areas	89.6	8.4	2.0
Donate for urgent needs	90.4	7.6	2.0
Overall Attitude level	**Good 73**	**Moderate 25**	**Poor 2**

Table 4 Association between demographic variables and blood donation practice ($n = 647$)

Variable	Practice of BD %		*p*-value
	Never donated	Ever donated	
Age (years)			
18–22	87.7	12.3	**< 0.001***
23–26	68.6	31.4	
27–30	53.1	46.9	
> 30	26.7	73.3	
Gender			
Male	73.2	26.8	**< 0.001***
Female	86.2	13.8	
Academic stream			
Health Sciences	76.2	23.8	**< 0.001***
Non-Health Sciences	87.9	12.1	
Year of study			
1st Year	86.5	13.5	**0.02***
2nd Year	78.6	21.4	
3rd Year	82.2	17.8	
4th Year	69.1	30.9	
Overall	80.5	19.5	

*Statistically significant at $p < 0.05$

3.3 Attitudes and perceptions of blood donation

Overall attitudes toward blood donation were highly positive, with most participants considering blood donation a noble social responsibility and expressing willingness to donate voluntarily. Most respondents also supported organizing regular blood donation camps in colleges and promoting donation activities in rural areas. Detailed attitude responses are summarized in Table 3.

3.4 Practice of blood donations

Despite positive attitudes toward blood donation, actual donation practice remained low. Blood donation practice was significantly associated with age, gender, academic stream, and year of study ($p < 0.05$). Donation practice was more common among older students, males, Health Sciences students, and senior-year students (Table 4).

3.5 Motivators and barriers of blood donation

The most commonly reported barriers to blood donation were lack of opportunity, medical reasons, and fear of needles. In contrast, major motivating factors included social

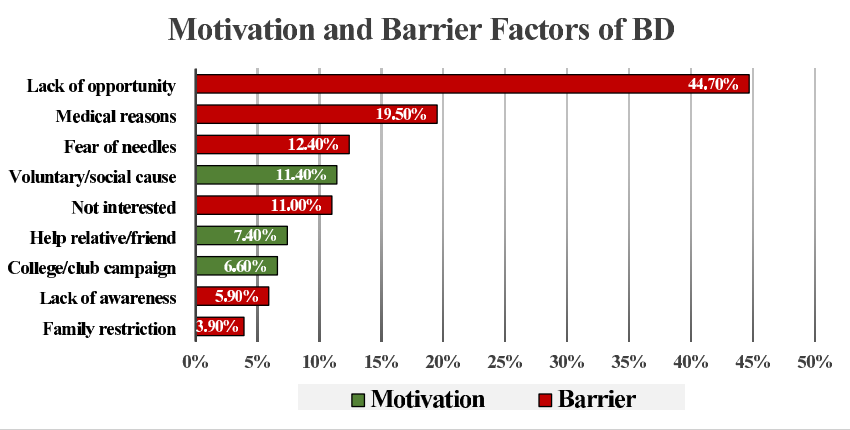


Fig. 2 Motivation and barrier factors of blood donation ($n = 647$)

Table 5 Sources of information about blood donation

Sources of information	%
TV/Radio	47.0
Newspapers/Magazines	32.8
Friends/Family	56.7
Social media	65.5
School/College programs	66.6
Blood donation camps	45.3
No information received	3.9

responsibility, helping family or friends, and participation in college or club campaigns. These findings suggest that limited access to donation opportunities, rather than unwillingness, is a major contributor to low donation practice among students (Fig. 2).

3.6 Sources of information about blood donation

The most common sources of information about blood donation were school or college awareness programs, social media, and friends or family. Many participants also reported learning about blood donation through blood donation camps and mass media, including television, radio, newspapers, and magazines. A small proportion of students reported not receiving any information regarding blood donation activities (Table 5). Multiple responses were allowed, percentages represent % of cases and may exceed 100%.

4 Discussion

This study evaluated the knowledge, attitude, and practice of blood donation among college students in Jumla, Nepal, and explored factors influencing donation behavior. A substantial knowledge-practice gap was observed: although students generally held positive attitudes, actual donation rates were low, consistent with reports from other developing regions [1, 4]. While most students showed positive attitudes and willingness to donate, actual donation practice remained low.

Overall knowledge of blood donation was low, with only 19.3% of participants achieving a good score, slightly lower than studies in Kathmandu (32.4%), reflecting the urban–rural disparity in awareness and access to blood donation information [12]. Health

Science students demonstrated higher awareness than non-health students, a trend also reported in South India and Kathmandu [11, 12]. However, technical knowledge-such as blood volume recovery and mandatory screening for transfusion-transmitted infections-was limited in both groups, consistent with a multicenter study across 16 countries, where only 28.5% had good knowledge [14]. These gaps suggest that while general awareness exists, detailed understanding of eligibility, procedures, and physiological recovery is lacking, highlighting the need for targeted awareness campaigns in Jumla.

Attitudes toward blood donation in Jumla were highly positive (73%), driven largely by humanitarian motives and moral satisfaction. Similar findings from Saudi Arabia and Qatar suggest that altruistic motivation remains a universal driver of voluntary blood donation across diverse cultural settings [10, 20]. Despite high willingness to donate (82.7%), many students had never been approached. The most common barrier was lack of opportunity (44.7%), a pattern observed globally, as a systematic review across nine countries found "not being asked" to be a leading reason for non-donation [9]. In the context of Jumla, "lack of opportunity" likely reflects limited local blood donation infrastructure and irregular availability of organized donation programs. Blood donation services in remote mountainous districts are often dependent on occasional hospital requests, temporary donation campaigns, or referrals to regional transfusion centers rather than permanently accessible blood bank facilities [16, 17]. Unlike urban areas where routine donation camps and transfusion centers are more available, students in Jumla may have limited exposure to regular donation events or convenient donation venues. Geographical barriers, transportation difficulties, and long travel distances to transfusion facilities may further discourage participation, particularly for first-time donors. At present, blood collection activities in the district are largely conducted through periodic institutional or emergency-based arrangements rather than routine campus-based voluntary donation systems. These contextual challenges may explain why willingness to donate was high despite low actual donation practice. Infrastructure limitations directly contribute to these barriers by reducing the availability, visibility, and accessibility of organized blood donation services. The absence of permanent blood collection centers and regular campus-based donation programs in Jumla limits opportunities for students to participate, even when they are willing. This gap could be addressed by establishing scheduled mobile blood donation camps, strengthening coordination between local institutions and the Nepal Red Cross Society, and integrating donation activities into routine college health programs [6].

The findings on sources of information indicate that students primarily relied on school/college programs, social media, and interpersonal networks, highlighting the importance of institutional and informal communication channels in shaping awareness. However, the presence of a small proportion of students reporting no exposure to blood donation information suggests gaps in structured awareness dissemination, which may contribute to the observed knowledge–practice gap.

Actual blood donation practice in Jumla was low (19.5%), compared to urban centers like Kathmandu, where replacement donations still constitute nearly 40% of the supply, indicating that VNRD is not yet the sole source [3, 8]. The region's scarcity of permanent transfusion centers creates a "blood desert" [11], and combined with misconceptions and fear associated with donation procedures, this further limits participation [7, 13]. Strengthening local transfusion infrastructure is essential to strengthen voluntary

blood donation practices in Jumla [8, 9]. Targeted educational interventions are needed to address fear of pain, a major barrier among 55% of students in similar settings [2], and innovative approaches, such as "walking blood banks" or mobile campus donation camps, can improve access in this remote region [11].

Several demographic factors were significantly associated with blood donation practice, including age, gender, academic stream, and year of study. Donation rates increased with age, possibly due to increased confidence, improved awareness regarding donation safety, and greater social engagement compared to younger students which is consistent with a multi-country study showing that maturity and social responsibility enhance voluntary participation [14]. Male students were more likely to donate than females (26.8% vs. 13.8%), reflecting physiological factors such as higher anemia prevalence among females and persistent misconceptions regarding menstruation and donation safety [1, 10, 12]. Furthermore, Health Sciences students demonstrated higher knowledge levels and greater participation in blood donation practices than Non-Health Sciences students, likely due to increased exposure to health education, disease prevention concepts, and clinical learning related to transfusion practices. Similarly, senior-year students, particularly those enrolled in Health Sciences programs, are more likely to encounter clinical environments, public health training, and emergency care settings where the importance of blood transfusion becomes more apparent and personally relevant. Comparable findings have also been reported in studies conducted in South Asia and other developing settings [2, 12, 21].

5 Strengths and limitations

This study provides the first baseline data on blood donation KAP in Jumla, a remote high-mountain district of Nepal. Inclusion of students from diverse academic streams - Health Sciences and Non-Health Sciences - enabled meaningful comparisons across educational backgrounds. The large sample size ($n = 647$) offered sufficient statistical power to identify demographic associations. However, the use of institution-based stratified convenience sampling may limit generalizability to all college students in Jumla. As a cross-sectional study, it captures only a snapshot in time and cannot establish causal relationships between academic progression and donation practices. Self-reported data may also introduce social desirability bias, particularly regarding attitudes. Although "medical reasons (underweight, anemia, illness)" were included as response options for non-donation, the study did not separately evaluate specific medical conditions such as anemia or chronic illness in detail. Additionally, although Jumla is a remote hilly region, the study was conducted in the main city area, so findings may differ in more isolated rural areas. Further, although the study was designed and conducted by pharmacy researchers, involvement of a practicing clinician or transfusion specialist in the interpretation of findings could have provided deeper clinical insight into transfusion service constraints and operational realities.

6 Recommendations

Based on the findings of this study, several strategies are recommended to strengthen voluntary blood donation in Jumla. Local authorities and health institutions should prioritize strengthening transfusion infrastructure and improving accessibility of blood donation services in remote regions to support a sustainable voluntary donation system.

Regular mobile blood donation camps, outreach activities, and campus-based donation programs should be organized in collaboration with the Nepal Red Cross Society and local healthcare institutions to address the major barrier of limited donation opportunities reported by students. In addition, regular blood donation awareness campaigns and college-based educational interventions should be integrated into student orientation programs and extracurricular activities to improve knowledge and convert willingness into actual donation practice. Digital awareness strategies and social media-based campaigns may further enhance student engagement and dissemination of donation-related information. Special attention should be given to Non-Health Sciences students and female students through targeted awareness initiatives aimed at addressing misconceptions, fear-related concerns, and gender-specific barriers to donation.

7 Conclusion

This study provides valuable insights into the knowledge, attitude, and practice of blood donation among college students in Jumla, Nepal. The findings indicate a strong foundation for voluntary donation, as most students demonstrated positive attitudes and high willingness to donate, perceiving it as a noble social responsibility. However actual donation practice remained low. Demographic factors such as age, gender, academic stream, and year of study significantly influenced donation behavior, with Health Science students and senior-year students showing higher engagement. Notably, lack of opportunity rather than unwillingness was the primary barrier, highlighting substantial untapped potential. Strengthening awareness programs and implementing regular campus-based blood donation campaigns may help translate willingness into sustained voluntary donation, thereby improving blood availability in remote regions such as Jumla.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12982-026-02862-6.

Supplementary Material 1.

Acknowledgements

The authors would like to express their sincere gratitude to the IRC-KAHS for providing ethical approval for this study. We are deeply thankful to the administrations and students of the KAHS, KTS, and JMC for their cooperation during the data collection process. Special thanks go to Anushesh Bhusal, Pawan Pandya, Bivek Yadav, Ritishna Duwal, Apekshya Sigdel, Nandani Gupta, Anjana Chaudhary and Sangita Sodari for needful help throughout the study. We also sincerely appreciate all the participants who voluntarily shared their time and insights for this research.

Author contributions

Deependra Muraw (DM): Data curation, formal analysis, investigation, writing – original draft preparation. Nisha Adhikari (NA): Data curation, formal analysis, writing – review & editing, supervision. Pradeep Narayan Joshi (PNJ): Data collection, investigation. Dirgha Raj Joshi (DRJ): Conceptualization, supervision, writing – review & final approval of the manuscript.

Funding

No funding was received for this work.

Data availability

The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

Ethics approval was obtained from the Institutional Review Committee (IRC) of the Karnali Academy of Health Sciences (KAHS) (Ref. No. 2025/084). All procedures involving human participants were conducted in accordance with institutional guidelines and regulations and complied with the ethical principles of the Declaration of Helsinki. Written informed consent was obtained from all participants prior to enrollment in the study. Participation was entirely voluntary. Participants were informed about the study objectives, procedures, and their right to refuse participation or withdraw at any time without any consequences.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Received: 8 March 2026 / Accepted: 25 August 2026



References

1. Singh S, Rijal TD, Katwal B. Blood donor's knowledge, attitude and practice towards blood donation among voluntary blood donors at Nepal red cross society central bood transfusion service Centre. J Nepalgunj Med Coll. 2023;21(2):5–9.
2. Kittu D, et al. Knowledge, attitude, and practice of blood donation among college students in Puducherry. Int J Med Sci Public Health. 2017;6(7):1231–7.
3. Neupane P et al. Knowledge, attitude and practice of blood donation among health science students at Kathmandu: a cross-sectional study. Nepal Med J. 2021;4(2).
4. Pokhrel B, Pandey N, Parajuli-Pokhrel K. Knowledge, attitude and practice regarding voluntary blood donation among the bachelor level students of Chitwan, Nepal. Janaki Med Coll J Med Sci. 2015;3(2):21–8.
5. Hada J. A research report on knowledge, attitude and practice on blood donation among 18–25 years bachelor level people. Hope International College. 2008:1–30.
6. World Health Organization. Blood safety and availability. 2025. https://www.who.int/news-room/fact-sheets/detail/blood-safety-and-availability. Accessed 11 May 2026.
7. Khan MS, et al. Knowledge, attitude, and practice of blood donation: a cross-sectional survey in Khulna city, Bangladesh. Public Health Pract. 2024;7:100488.
8. Gurung N, et al. Comparison of volunteer non-remunerated donors and replacement donors. J Nepal Health Res Counc. 2022;20(04):973–6.
9. Easmin F, Islam MA. Assessment of knowledge, attitude and practice of donors towards blood donation and outcome of an intervention to change the behavior of donors-a systemic review. Saudi J Med Pharm Sci. 2024;10(4):231–7.
10. Ibrahim M, et al. Knowledge level, motivational factors, and potential barriers toward blood donation among MBBS students. Students' Supplement J Rawalpindi Med Coll. 2023;27(S–1):75–82.
11. Raykar NP, et al. Innovative blood transfusion strategies to address global blood deserts: a consensus statement from the blood delivery via emerging strategies for emergency remote transfusion (Blood DESERT) coalition. Lancet Global Health. 2024;12(3):e522–9.
12. Amatya M. Study on knowledge, attitude and practice of blood donation among students of different colleges of Kathmandu, Nepal. Int J Pharm Biol Archives. 2013;4(3):424–8.
13. Hossain MS et al. Knowledge, attitude, and practice towards blood donation among residential students and teachers of religious institutions in Bangladesh–A cross-sectional study. Heliyon. 2022;8(10):e410792.
14. Eltewacy NK, et al. Unveiling blood donation knowledge, attitude, and practices among 12,606 university students: a cross-sectional study across 16 countries. Sci Rep. 2024;14(1):8219.
15. Sabu K et al. Knowledge, attitude and practice on blood donation among health science students in a university campus, South India. Online J Health Allied Sci. 2011;10(2).
16. Paudel P, Nepal B. Blood transfusion services in Nepal: an overview. Asian J Transfus Sci. 2025;20(1);104103.
17. Jha KK, Rajput PS. Blood bank service: an experience at Karnali Academy of Health Sciences, Jumla, Nepal. MedS Alliance J Med Med Sci. 2021;1(2):87–9.
18. Salem M, et al. Knowledge, attitude, and practice of blood donation among undergraduate medical students in Syria. Medicine. 2024;103(8):e37086.
19. Kaliyaperumal K. Guideline for conducting a knowledge, attitude and practice (KAP) study. AECS illumination. 2004;4(1):7.
20. Mahfouz MS, et al. Blood donation among university students: practices, motivations, and barriers in Saudi Arabia. Avicenna J Med. 2021;11(02):70–6.
21. Mamatya A, Prajapati R, Yadav R. Knowledge and practice of blood donation: a comparison between medical and non-medical Nepalese students. Nepal Med Coll J. 2012;14(4):283–6.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.